

Sonometer

1 Aim

1. To observe standing wave modes in a stretched string.
2. To vary tension, length and linear density of the sonometer wire and observe changes in frequency of vibrated string.
3. To find the linear density of an unknown wire.

2 Introduction

A sonometer is a hollow box with two fixed wedges near the ends and a pulley fixed at one end. A string (metal wire) is fixed at one end, run over the bridges and the pulley and attached to a weight holder hanging below the pulley. Weights can be added to the hanger to vary the tension in the wire and a third movable bridge can be placed to change the length of the vibrating string.

The sonometer shows the relationship between the frequency of sound produced by a plucked string, the tension, length and mass per unit length of the string. The frequency of mode n of a perfect string fixed at the ends is given by

$$f_n = n \frac{\nu}{2L}$$

where L is the string length and f_n are the frequencies at which the string will resonate. The lowest (the fundamental) node has $n = 1$. When transverse waves are excited in the stretched wire of the sonometer, the wedges reflect these waves. In the fundamental mode, the wavelength λ of the waves in the string is given by $\lambda = 2l$ where l is the distance between the bridges.

3 Apparatus in the lab and procedure

1. Place a load in the hanger, and check that the wire is stretched between the wedges and touches them.



Figure 1: Apparatus in the Lab.

2. Keep the wedges close to each other. Cut a small piece of paper and fold it so that it can be placed on the wire.
3. Take one of the tuning forks and strike it on the rubber pad and set vibrations. Press the stem of the tuning fork on the sonometer board.
4. vary distance between the wedges (very slowly) while keeping the paper rider in the centre and continue till the rider flies off. The wire is now vibrating in resonance with the tuning fork frequency.
5. Measure the length between the wedges using the meter scale provided to you. Increase the distance between the wedges and adjust the length for resonance again.
6. Repeat the above steps four/five times.

3.1 Precautions

1. Make sure there are not kinks in the wire.
2. Do not forget to weigh the hanger and include it in your calculations.
3. Make sure that the pulley has at little friction as possible.